

Kumardeep Singh

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SUMMARY

- Fourth-year Bachelor of Technology in Information Technology student at KPU (GPA: 3.98, Dean's Honor Roll) with hands-on experience designing and building full-stack applications, RESTful APIs, and desktop systems using Java, Spring Boot, React, PHP, Python, and SQL.
- Proficient in Java, Python, JavaScript, TypeScript, and SQL; experienced applying OOP design patterns, MVC/layered architecture, RESTful API design, OAuth2/JWT security, Docker, and Git across academic and personal projects.
- Skilled in full-stack web development, database design, third-party API integration, and software quality assurance; experienced using Spring Boot, React, PostgreSQL, and MySQL to build production-quality applications.
- Strong analytical thinker and collaborative communicator with proven ability to design, build, document, and deliver complete software solutions from requirements through deployment.

EDUCATION

KWANTLEN POLYTECHNIC UNIVERSITY

Bachelor of Technology in Information Technology

Cumulative GPA: 3.89; Dean's Honor roll Fall 2025

Surrey, BC

Expected Dec 2026

Relevant Coursework:

- **Object-Oriented Software Engineering:** Applied UML, SOLID principles, design patterns, and Agile methodologies to analyze, design, and implement object-oriented software systems; produced system models and design documentation from requirements.
- **Database Systems:** Gained proficiency in relational database design, entity-relationship modeling, and normalization; wrote SQL queries to design, maintain, and query databases, and integrated Java and Python applications with databases using JDBC and Python APIs.
- **Web Application Development:** Developed and deployed full-stack data-driven web applications using HTML, CSS, JavaScript, and server-side programming; applied client-server architecture, REST principles, and web hosting concepts to build interactive, functional platforms.

PROJECTS

Event Ticket Platform

Technologies: Java 21, Spring Boot,

React, TypeScript, PostgreSQL, Docker, Keycloak, Maven

- Built a full-stack event ticketing application using Java 21 and Spring Boot for the backend and React with TypeScript for the frontend, with PostgreSQL managed via Spring Data JPA and Maven.
- Designed and implemented RESTful APIs using a layered Controller-Service-Repository architecture, applying Spring Security with OAuth2 and JWT-based authentication via Keycloak for role-based access control.
- Applied **MVC architecture** and multi-step input validation to maintain accurate and consistent records across all user roles. Applied **SOLID principles** and OOP design patterns for clean, maintainable architecture.
- Integrated **JDBC** for database interaction with structured data modeling and normalization to maintain accurate records across all user roles.
- Documented system architecture, security configurations, and data flows; tracked and resolved application issues through structured debugging workflows.
- Gathered and documented system requirements and data models to support maintainability and team reference.

PeerTutor

Technologies: HTML, CSS, JavaScript, PHP, MySQL, Zoom API, Stripe API

- Built a full-stack peer tutoring web application enabling students to find and book verified tutors based on subject and availability, with role-based interfaces for students, tutors, and admins.
- Implemented user registration, tutor skill verification, and session scheduling with approval logic; built an admin dashboard and matching algorithm for efficient tutor-student pairing.
- Followed **MVC pattern** and **object-oriented design** for clean code separation.
- Implemented **CRUD operations**; designed a responsive, multi-role interface and handled backend logic, session management, authentication flows, and persistent data operations with PHP and MySQL.
- Conducted functional and QA testing across all user roles to validate data integrity and system behaviour; maintained source code with Git and produced documentation covering system components and API integrations.

Blog Application

Technologies: Java, Spring Boot, React,

TypeScript, PostgreSQL, JWT, Spring Security

- Built a full-stack web application using Spring Boot for backend services and React with TypeScript for the frontend, applying Domain-Driven Design and Onion Architecture principles.
- Designed and implemented RESTful APIs using a Controller-Service-Repository layered architecture; implemented authentication and authorization with Spring Security and JWT for role-based access control.
- Integrated **Zoom API** for automatic meeting link generation and **Stripe** for session-based payments with platform commission.
- Developed reusable frontend components with clean, decoupled API service layers and unidirectional data flow for single-source-of-truth state management.
- Used **PHP and MySQL** for backend logic; emphasized code quality, peer-review readiness, and maintainability in alignment with established software development standards.

Technical Skills

Programming Languages: Java, Python, JavaScript, TypeScript, PHP, SQL, HTML/CSS

Frameworks & Libraries: Spring Boot, Spring Security, React, Docker, REST APIs, OAuth2, JWT, JDBC, Maven

Tools: Git, GitHub, Linux, MySQL, PostgreSQL, Command Line Interface